Operation Management

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<MBAZG526/QMZG526, Operation Management>
Lecture No. 11

Inventory Models

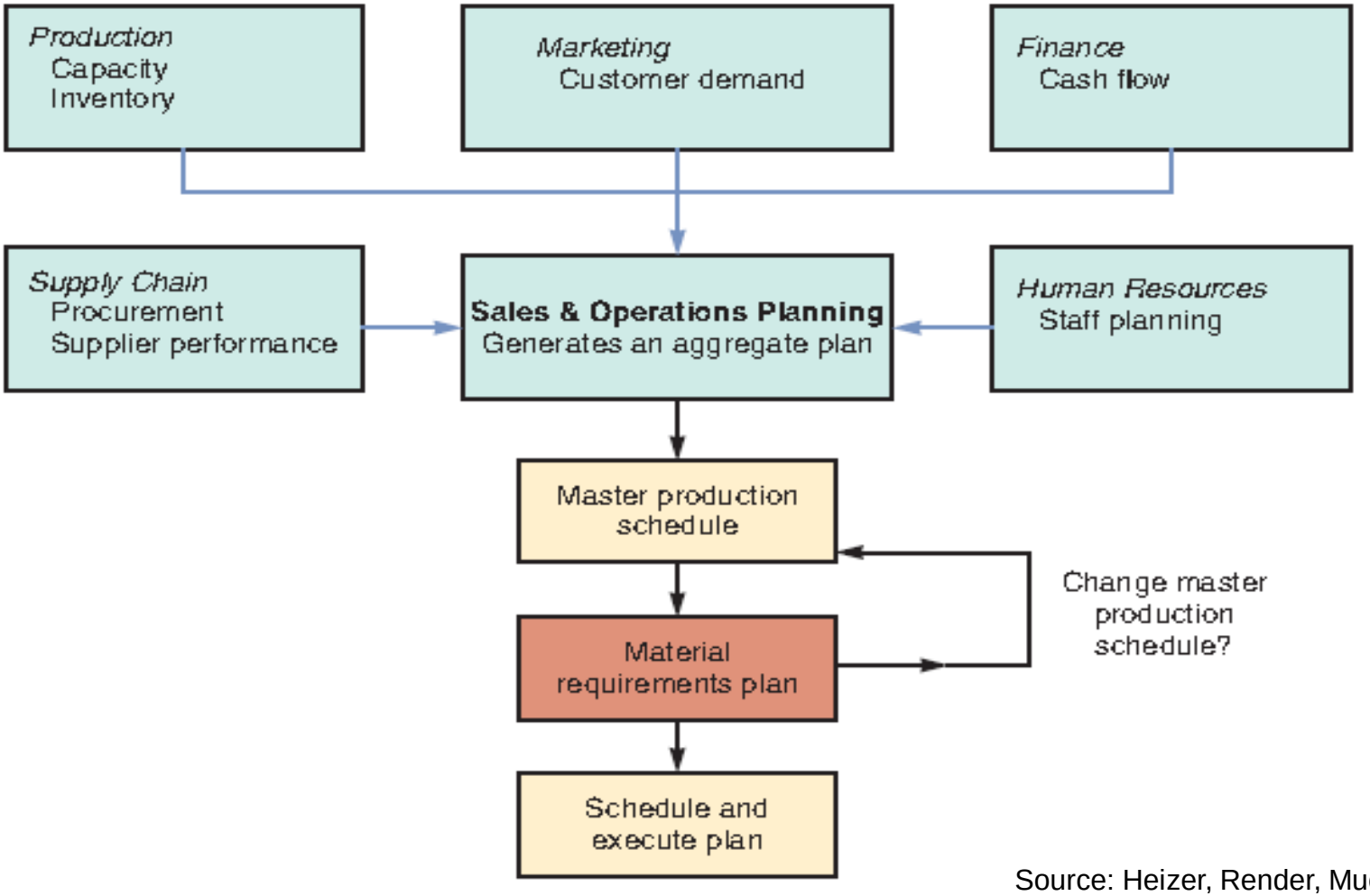
- Independent Demand Model.
- Dependent Demand Model. (Will be deal later in Dependent Demand Management)

Dependent Demand

- Dependent demand" means the demand for one item is related to the demand for another item.

MRP

- **Materials Requirement planning (MRP)** : A dependent demand technique that uses a bill-of-material, inventory, expected receipts, and a master production schedule to determine material requirements.



Source: Heizer, Render, Muckson

Dependent Demand

- Benefits of MRP

1) Better response to customer orders
2) Faster response to market changes
3) Improved utilization of facilities and labor
4) Reduced inventory levels

IPO of MRP

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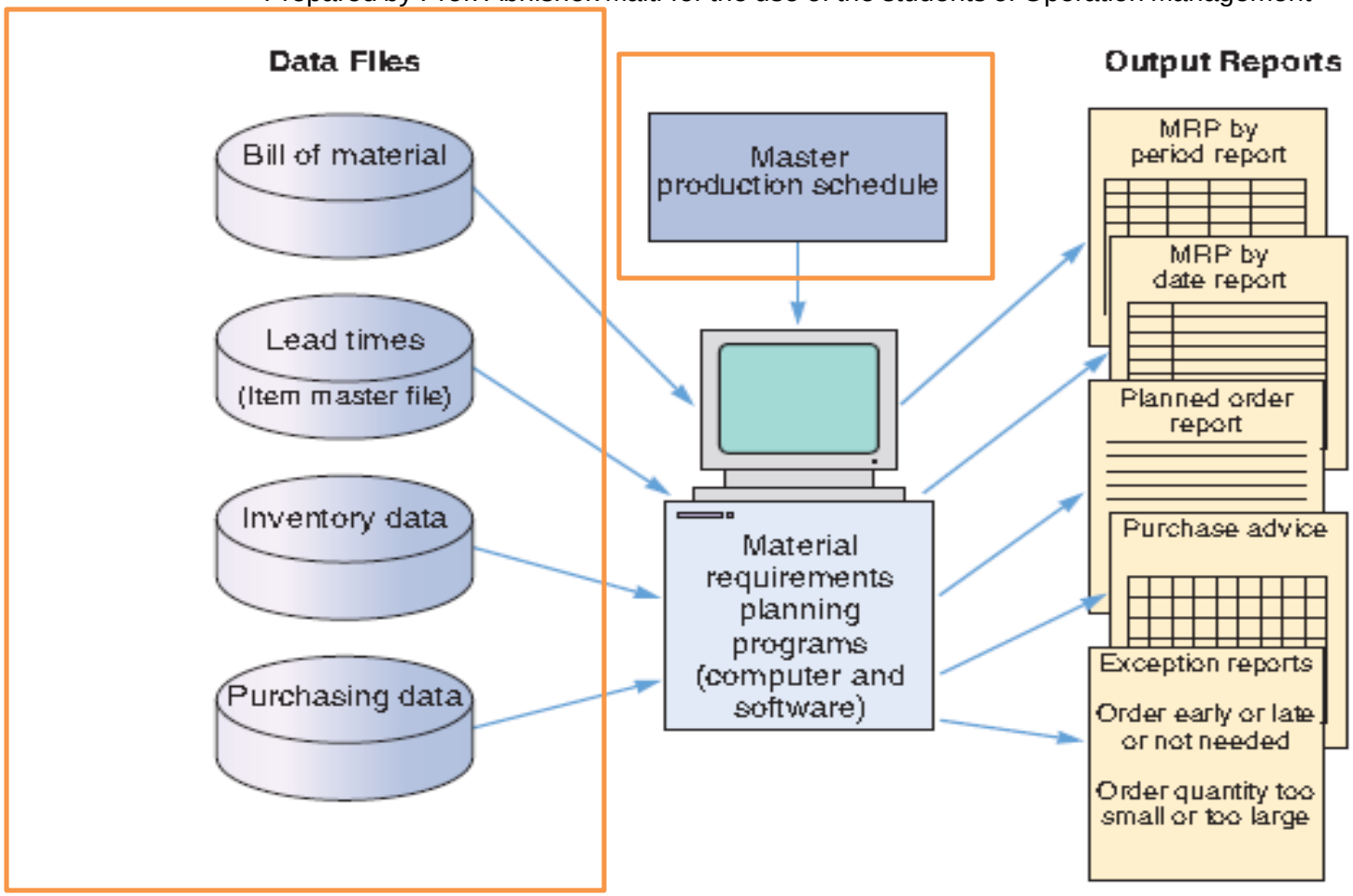

Source: Heizer, Render, Muckson

Master Production Schedule

- Master production schedule (MPS) : A timetable that specifies what is to be made (usually finished goods) and when.

Aggregate Plan
(Shows the total quantity of amplifiers)

Months	January	February
	1,500	1,200

Master Production Schedule
(Shows the specific type and quantity of amplifier to be produced)

Weeks	1	2	3	4	5	6	7	8
240-watt amplifier	100		100		100		100	
150-watt amplifier		500		500		450		450
75-watt amplifier			300				100	

Source: Heizer, Render, Muckson

Bill of Materials (BOM)

- Bill of material (BOM) : A listing of the raw materials, parts, components, assembly, sub-assembly, their description, and the quantity of each required to make one unit of a product.
- Product structure tree : A visual depiction of the requirements in a bill of materials, where all components are listed by levels.

Bill of Materials (BOM)

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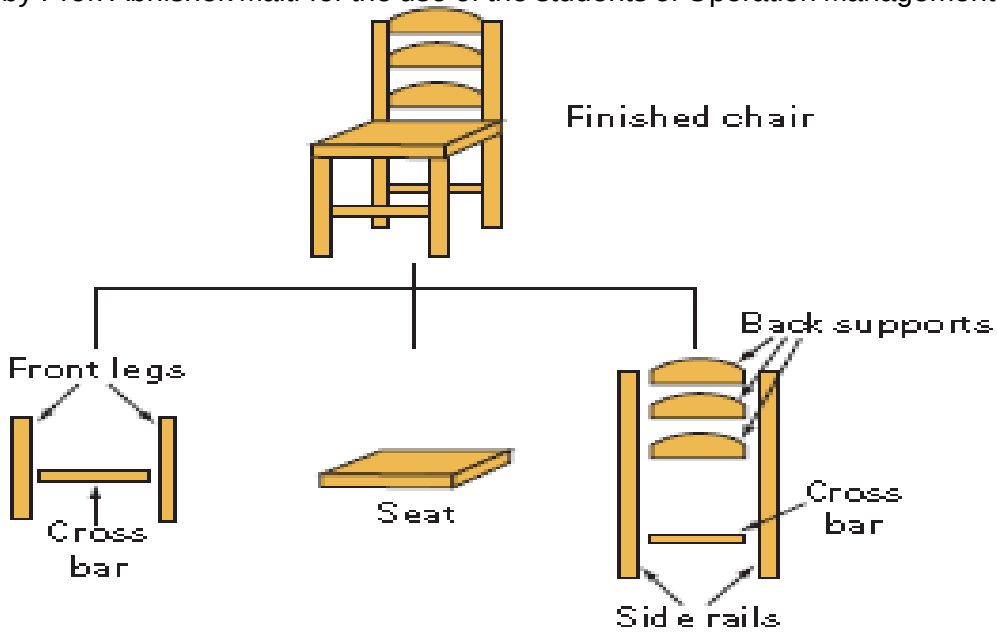
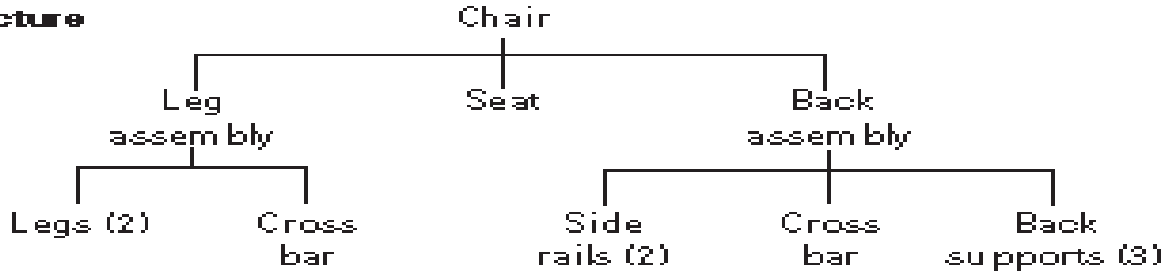
Assembly diagram



Product structure tree



Source: Operation Management by William J Stevenson

Bill of Materials (BOM)

- Modular Bills : Bills of material organized by major subassemblies or by product options.
- Planning bills : Material groupings created in order to assign an artificial parent to a bill of material; also called “pseudo” bills.
- Phantom bills : Bills of material for components, that exist only temporarily; they are never inventoried.

Bill of Materials (BOM)

- Low-level coding : Restructuring the bill of materials so that multiple occurrences of a component all coincide.

Inventory Records

- Inventory records: It includes information on the status of each item by time period.

Purchasing Data

- Knowledge of outstanding orders exists as a by-product of well-managed purchasing and inventory-control departments is known as **purchasing order outstanding.**
- Lead time : In purchasing systems, the time between recognition of the need for an order and receiving it; in production systems, it is the order, wait, move, queue, setup, and run times for each component.

IPO of MRP

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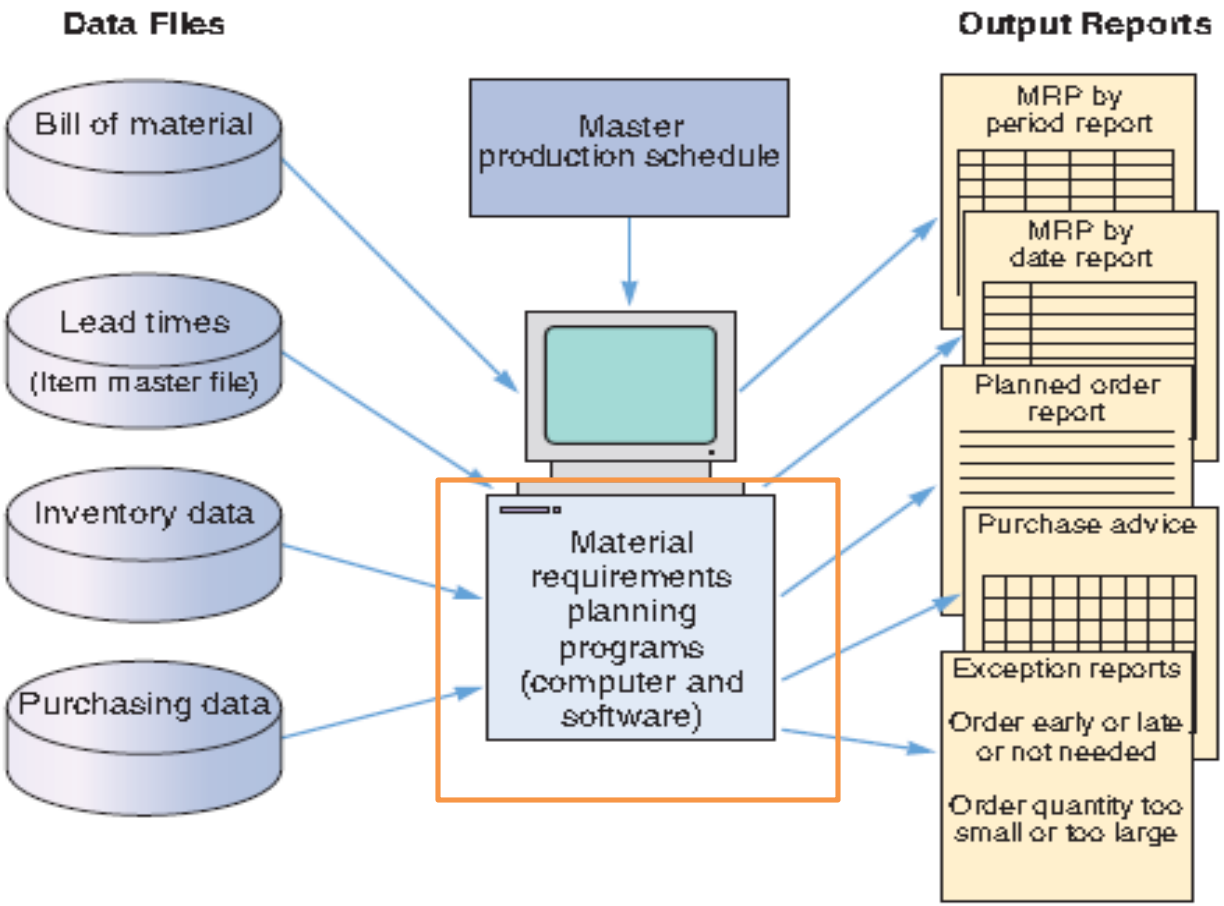


Source: Heizer, Render, Muckson

Material Requirements Planning (MRP)

Aspect	Aggregate Planning	Master Production Scheduling (MPS)
Goal	Focuses on overall production strategy	Focuses on manufacturing details of a specific product
Target	Balance production capacity with market demands	Create a roadmap for production, ensuring quality and on-time delivery
Flexibility	Adapts easily to changing market conditions	Is detailed and frequent changes can disturb production flow

Terminology

- Gross requirements : Total expected demand for an item or raw material in a time period.
- Scheduled receipts : Open orders scheduled to arrive from vendors or elsewhere in the pipeline.
- Projected on hand : Expected amount of inventory that will be on hand at the beginning of each time period.
- Net requirements : The actual amount needed in each time period.

Terminology

- Planned-order receipts : Quantity expected to be received by the beginning of the period in which it is shown.
- Planned-order releases : Planned amount to order in each time period.

Formula

- The quantities that are generated by exploding the bill of materials are *gross requirements.*

Net requirements for current period = Gross requirements for current period − Projected on-hand inventory for current period

Project inventory on-hand for current period = Planned receipts for previous period − Net requirements for previous period + Scheduled receipts for current period

Class Exercise

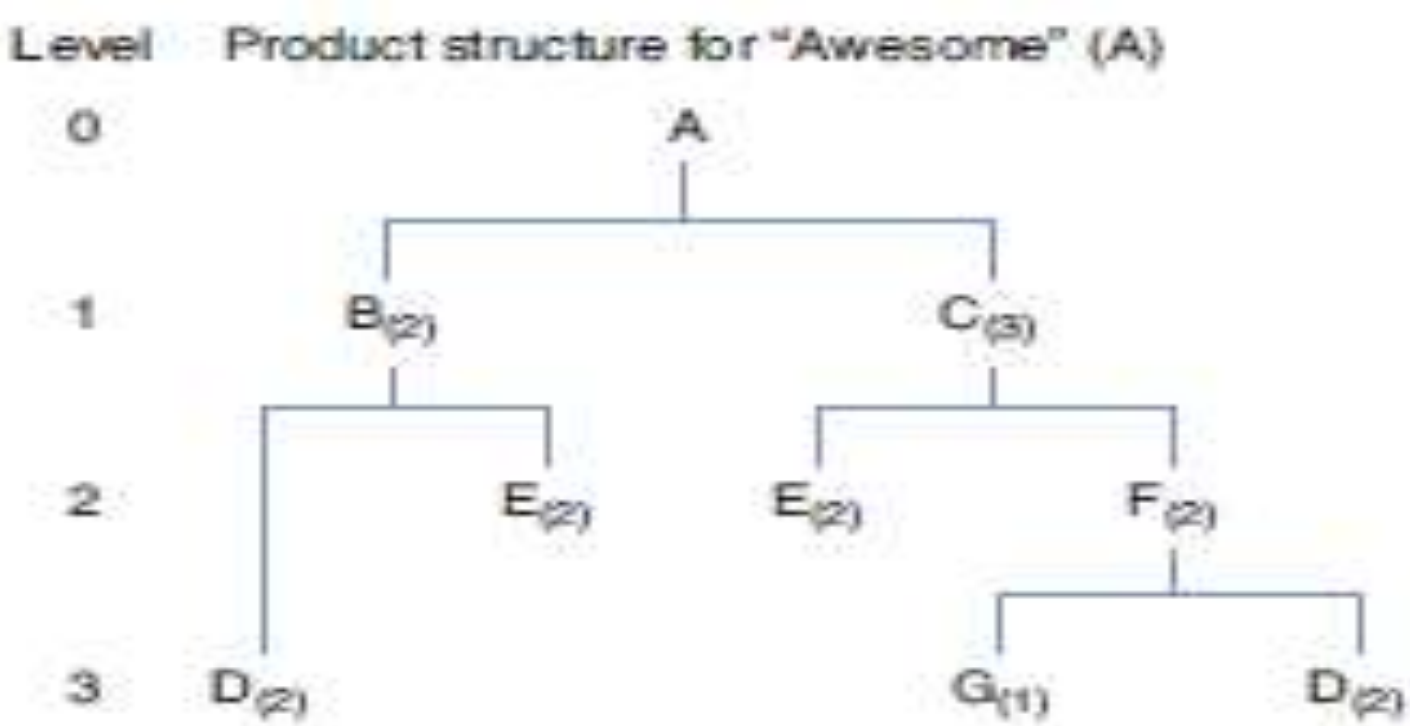
DEVELOPING A PRODUCT STRUCTURE AND GROSS REQUIREMENTS

- Speaker Kits, Inc., packages high-fidelity components for mail order.
- Components for the top-of-the line speaker kit, “Awesome” (A), include 2 Bs and 3 Cs.
- Each B consists of 2 Ds and 2 Es.
- Each of the Cs has 2 Fs and 2 Es.

Class Exercise

- Each F includes 2 Ds and 1 G.
- The demand for 50 Awesome Speaker Kits.

Solution



Source: Heizer, Render, Muckson

Solution

Once we have developed the product structure, we can determine the number of units of each item required to satisfy demand for a new order of 50 Awesome speaker kits. We "explode" the requirements as shown:

Part B:	2 × number of As =	(2)(50) =	100
Part C:	3 × number of As =	(3)(50) =	150
Part D:	2 × number of Bs + 2 × number of Fs =	(2)(100) + (2)(300) =	800
Part E:	2 × number of Bs + 2 × number of Cs =	(2)(100) + (2)(150) =	500
Part F:	2 × number of Cs =	(2)(150) =	300
Part G:	1 × number of Fs =	(1)(300) =	300

Source: Heizer, Render, Muckson

Class Exercise

BUILDING A GROSS REQUIREMENTS PLAN

- Each Awesome speaker kit requires all the items in the product structure for A. Lead times are shown in Table.
- We construct the gross material requirements plan with a production schedule that will satisfy the demand of 50 units of A by week 8.
- Assembly and Production.

TABLE 14.3 Gross Material Requirements Plan for 50 Awesome Speaker Kits (As) with Order Release Dates Also Shown

	WEEK								LEAD TIME
	1	2	3	4	5	6	7	8	
A. Required date								50	
Order release date							50		1 week
B. Required date							100		
Order release date					100				2 weeks
C. Required date							150		
Order release date						150			1 week
E. Required date					200	300			
Order release date			200	300					2 weeks
F. Required date						300			
Order release date			300						3 weeks
D. Required date			600		200				
Order release date		600		200					1 week
G. Required date			300						
Order release date	300								2 weeks

Source: Heizer, Render, Muckson

Solution

- If you want 50 units of A at week 8, you must start assembling A in week 7.
- Thus, in week 7, you will need 100 units of B and 150 units of C. These two items take 2 weeks and 1 week, respectively, to produce.
- Production of B, therefore, should start in week 5, and production of C should start in week 6.

Class Exercise

DETERMINING NET REQUIREMENTS PLAN

- Speaker Kits, Inc., now wants to construct a net requirements plan.
- The gross requirement remains 50 units in week 8, and component requirements are as shown in the product structure

ITEM	ON HAND	ITEM	ON HAND
A	10	E	10
B	15	F	5
C	20	G	0
D	10		

Source: Heizer, Render, Muckson

Solution

Net Material Requirements Plan for 50 Units of Product A in Week 8. *(The superscript is the source of the demand)*

Lot Size	Lead Time (weeks)	On Hand	Safety Stock	Allo-cated	Low-Level Code	Item Identi-fication			Week 1	2	3	4	5	6	7	8
Lot-for-Lot	1	10	—	—	0	A	Gross Requirements									50
							Scheduled Receipts									
							Projected On Hand	10	10	10	10	10	10	10	10	10
							Net Requirements									40
							Planned Order Receipts									40
							Planned Order Releases								40	
Lot-for-Lot	2	15	—	—	1	B	Gross Requirements								80^A	
							Scheduled Receipts									
							Projected On Hand	15	15	15	15	15	15	15	15	
							Net Requirements								65	
							Planned Order Receipts								65	
							Planned Order Releases					65				
Lot-for-Lot	1	20	—	—	1	C	Gross Requirements								120^A	
							Scheduled Receipts									
							Projected On Hand	20	20	20	20	20	20	20	20	
							Net Requirements								100	
							Planned Order Receipts								100	
							Planned Order Releases						100			
Lot-for-Lot	2	10	—	—	2	E	Gross Requirements						130^B	200^C		
							Scheduled Receipts									
							Projected On Hand	10	10	10	10	10	10			
							Net Requirements						120	200		
							Planned Order Receipts						120	200		
							Planned Order Releases			120	200					
Lot-for-Lot	3	5	—	—	2	F	Gross Requirements							200^C		
							Scheduled Receipts									
							Projected On Hand	5	5	5	5	5	5	5		
							Net Requirements							195		
							Planned Order Receipts							195		
							Planned Order Releases			195						

$2 \times$ number of As = 80

$3 \times$ number of As = 120

$2 \times$ number of Bs = 130

$2 \times$ number of Cs = 200

$2 \times$ number of Cs = 200

Source: Heizer, Render, Muckson

Solution

Lot-for-Lot	1	10	—	—	3	D	Gross Requirements				390[F]		130[B]			
							Scheduled Receipts									
							Projected On Hand	10	10	10	10					
							Net Requirements				380		130			
							Planned Order Receipts				380		130			
							Planned Order Releases			380		130				

2 × number of Bs = 130

2 × number of Fs = 390

Lot-for-Lot	2	0	—	—	3	G	Gross Requirements				195[F]					
							Scheduled Receipts									
							Projected On Hand				0					
							Net Requirements				195					
							Planned Order Receipts				195					
							Planned Order Releases		195							

1 × number of Fs = 195

Source: Heizer, Render, Muckson

Any Question

Safety Stock

- BOMs, inventory records, purchase and production quantities may not be perfect
- Consideration of safety stock may be prudent
- Should be minimized and ultimately eliminated
- Typically built into projected on-hand inventory

MRP Management

- **MRP dynamics**
  - Facilitates replanning when changes occur
  - **System nervousness** can result from too many changes
  - **Time fences** put limits on replanning
  - **Pegging** links each item to its parent allowing effective analysis of changes

MRP Management

- **MRP limitations**
  - MRP does not do detailed scheduling–it plans
  - Works best in product-focused, repetitive environments
  - Requires fixed lead time and infinite size time **buckets**

- Cannot be applicable in Make-to-Order environment.
- MRP is considered an *infinite* scheduling technique.

Lot Sizing Techniques

- Lot for Lot : A lot-sizing technique that generates exactly what is required to meet the plan.
- Lot Sizing with EOQ : A lot-sizing technique that generates Economic Order Quantity.
- Lot Sizing for POQ : An inventory ordering technique that issues orders on a predetermined time interval, with the order quantity covering the total of the interval's requirements.

Lot-Sizing Techniques

- **Lot-for-lot technique orders just what is required for production based on net requirements**
  - May not always be feasible
  - If setup costs are high, lot-for-lot can be expensive
- **Economic order quantity (EOQ)**
  - EOQ expects a known constant demand and MRP systems often deal with unknown and variable demand

Lot-Sizing Techniques

- **Periodic order quantity (POQ) orders quantity needed for a predetermined time period**
  - Interval = EOQ / average demand per period
  - Order quantity set to cover the interval
  - *Order quantity recalculated at the time of the order release*
  - No extra inventory

Lot-Sizing Techniques

- ***Dynamic lot sizing* techniques**
  - Balance lot size and setup costs
  - *Part period balancing* (*least total cost*)
  - *Least unit cost*
  - *Least period* cost (*Silver-Meal*)
- **Dynamic programming approach**
  - *Wagner-Whitin*

Lot for Lot

MRP Lot Sizing: Lot-for-Lot Technique*

WEEK		1	2	3	4	5	6	7	8	9	10
Gross requirements		35	30	40	0	10	40	30	0	30	55
Scheduled receipts											
Projected on hand	35	35	0	0	0	0	0	0	0	0	0
Net requirements		0	30	40	0	10	40	30	0	30	55
Planned order receipts			30	40		10	40	30		30	55
Planned order releases		30	40		10	40	30		30	55	

*Holding costs = $1/unit/week; setup cost = $100; gross requirements average per week = 27; lead time = 1 week.

Source: Heizer, Render, Muckson

Solution

- The holding cost is zero as there is never any end-of-period inventory.
- But seven separate setups (one associated with each order) yield a total cost of
- \$700. (Holding cost = 0 × 1 = 0; ordering cost = 7 × 100 = 700.)

Lot Sizing with EOQ

- With a setup cost of $100 and a holding cost per week of $1, Speaker Kits, Inc., wants to examine its cost with lot sizes based on an EOQ criteria.

$$Q^* = \sqrt{\frac{2DS}{H}}$$

D = annual usage = 1,404

S = setup cost = $100

H = holding (carrying) cost, on an annual basis per unit

= $1 × 52 weeks = $52

$$Q^* = 73 \text{ units}$$

Source: Heizer, Render, Muckson

Lot Sizing with EOQ

Therefore, place an order of 73 units, as necessary, to avoid a stockout.

MRP Lot Sizing: EOQ Technique*

WEEK		1	2	3	4	5	6	7	8	9	10
Gross requirements		35	30	40	0	10	40	30	0	30	55
Scheduled receipts											
Projected on hand	35	35	0	43	3	3	66	26	69	69	39
Net requirements		0	30	0	0	7	0	4	0	0	16
Planned order receipts			73			73		73			73
Planned order releases		73			73		73			73	

*Holding costs = \$1/unit/week; setup cost = \$100; gross requirements average per week = 27; lead time = 1 week.

For the 10-week planning period:

Holding cost = 375 units × \$1 = \$375 (includes 57 remaining at the end of week 10)

Ordering cost = 4 × \$100 = \$400

Total = \$375 + \$400 = \$775

Source: Heizer, Render, Muckson

Lot Sizing with POQ

- With a setup cost of $100 and a holding cost per week of $1, Speaker Kits, Inc., wants to examine its cost with lot sizes based on an POQ criteria.

Lot Sizing with POQ

- Ten-week usage equals a gross requirement of 270 units; therefore, average weekly usage equals 27, and, we know the EOQ is 73 units.
- Therefore:
- POQ interval = EOQ/Average weekly usage = 73/27 = 2.7, or 3 weeks.

MRP Lot Sizing: POQ Technique*

WEEK		1	2	3	4	5	6	7	8	9	10
Gross requirements		35	30	40	0	10	40	30	0	30	55
Scheduled receipts											
Projected on hand	35	35	0	40	0	0	70	30	0	0	55
Net requirements		0	30	0	0	10	0	0	0	55	0
Planned order receipts			70			80		0		85	0
Planned order releases		70			80				85		

*Holding costs = \$1/unit/week; setup cost = \$100; gross requirements average per week = 27; lead time = 1 week.

$$\text{Setups} = 3 \times \$100 = \$300$$

$$\text{Holding cost} = (40 + 70 + 30 + 55)\ \text{units} \times \$1\ \text{each} = \$195$$

The POQ solution yields a computed 10-week cost of \$300 + \$195 = \$495

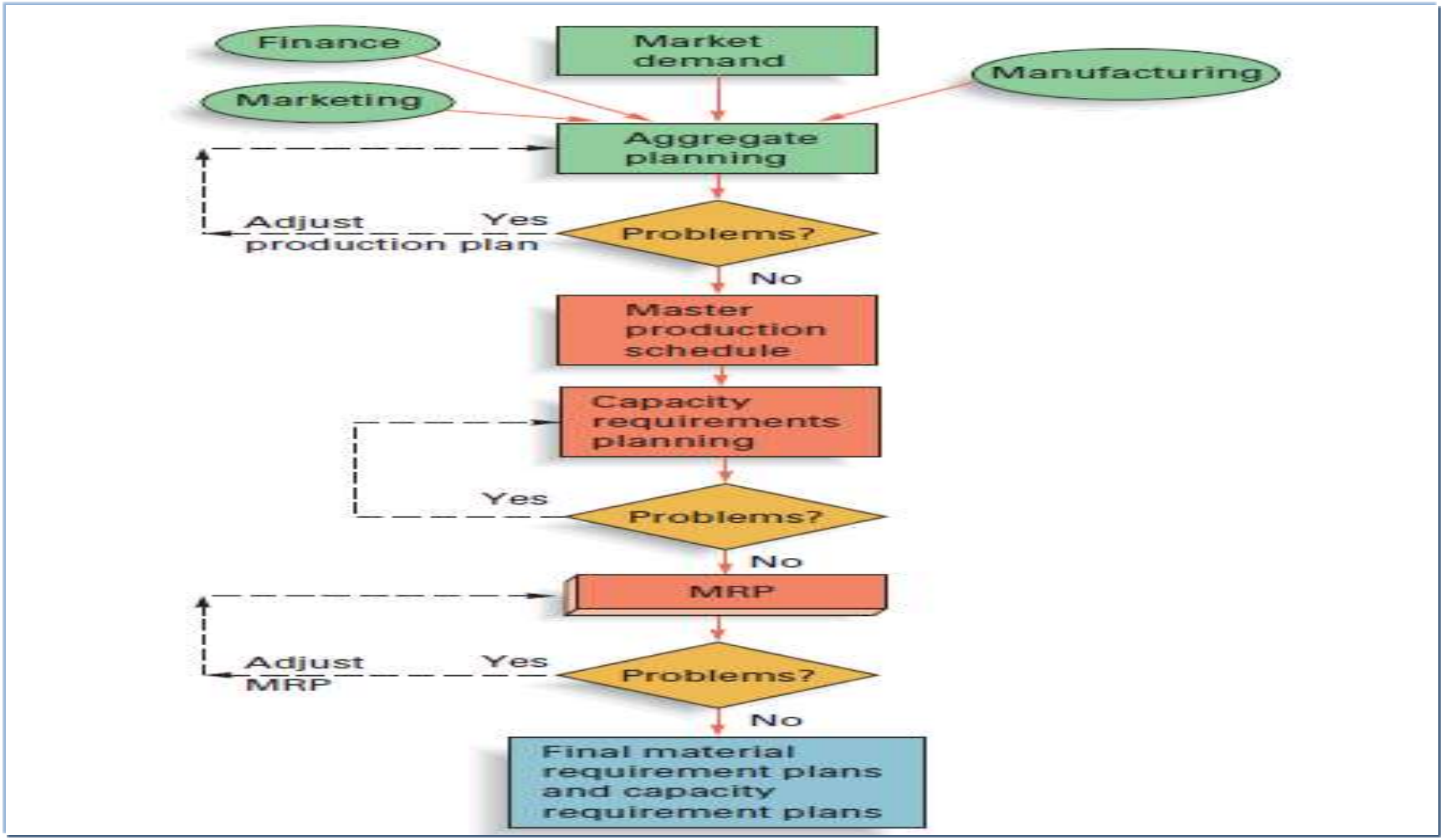
Source: Heizer, Render, Muckson

Comparison

	COSTS		
	SETUP	HOLDING	TOTAL
Lot-for-lot	$700	$0	$700
Economic order quantity (EOQ)	$400	$375	$775
Periodic order quantity (POQ)	$300	$195	$495

Source: Heizer, Render, Muckson

MRP-II



Source: Operation Management by William J Stevenson

Material Requirements Planning II

- Requirement data can be enriched by other resources
- Generally called MRP II or Material Resource Planning
- Outputs can include scrap, packaging waste, effluent, carbon emissions

- Data used by purchasing, production scheduling, capacity planning, inventory, warehouse management

Closed Loop MRP

- A system that provides feedback to the capacity plan, master production schedule, and production plan so planning can be kept valid at all times.
- Load report : A report showing the resource requirements in a work center for all work currently assigned there as well as all planned and expected orders.

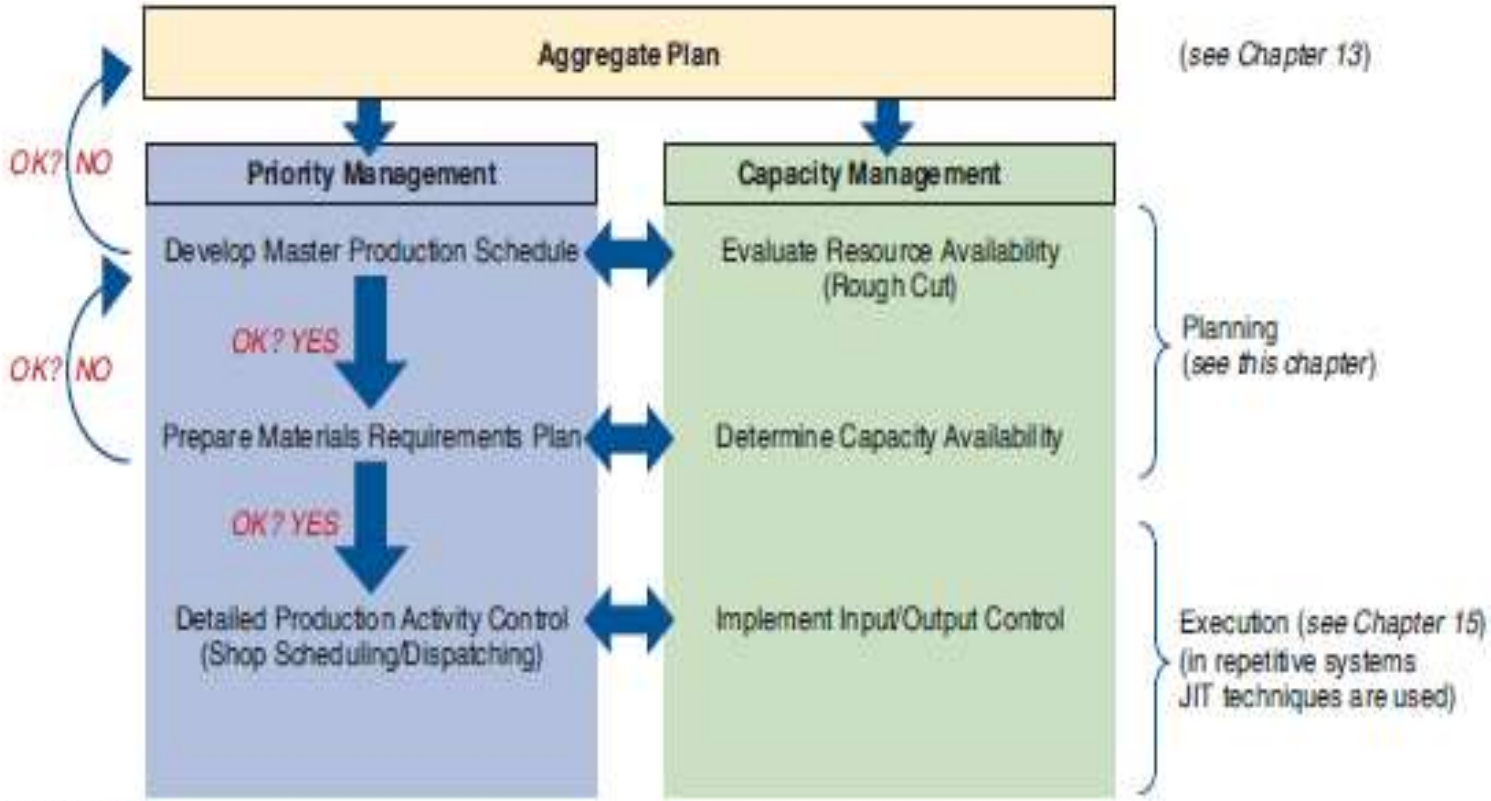


Figure 14.7

Closed-Loop Material Requirements Planning

Capacity Planning

- Feedback from the MRP system
- Load reports show resource requirements for work centers
- Work can be moved between time periods or work centers to smooth the load or bring it within capacity

Smoothing the Load

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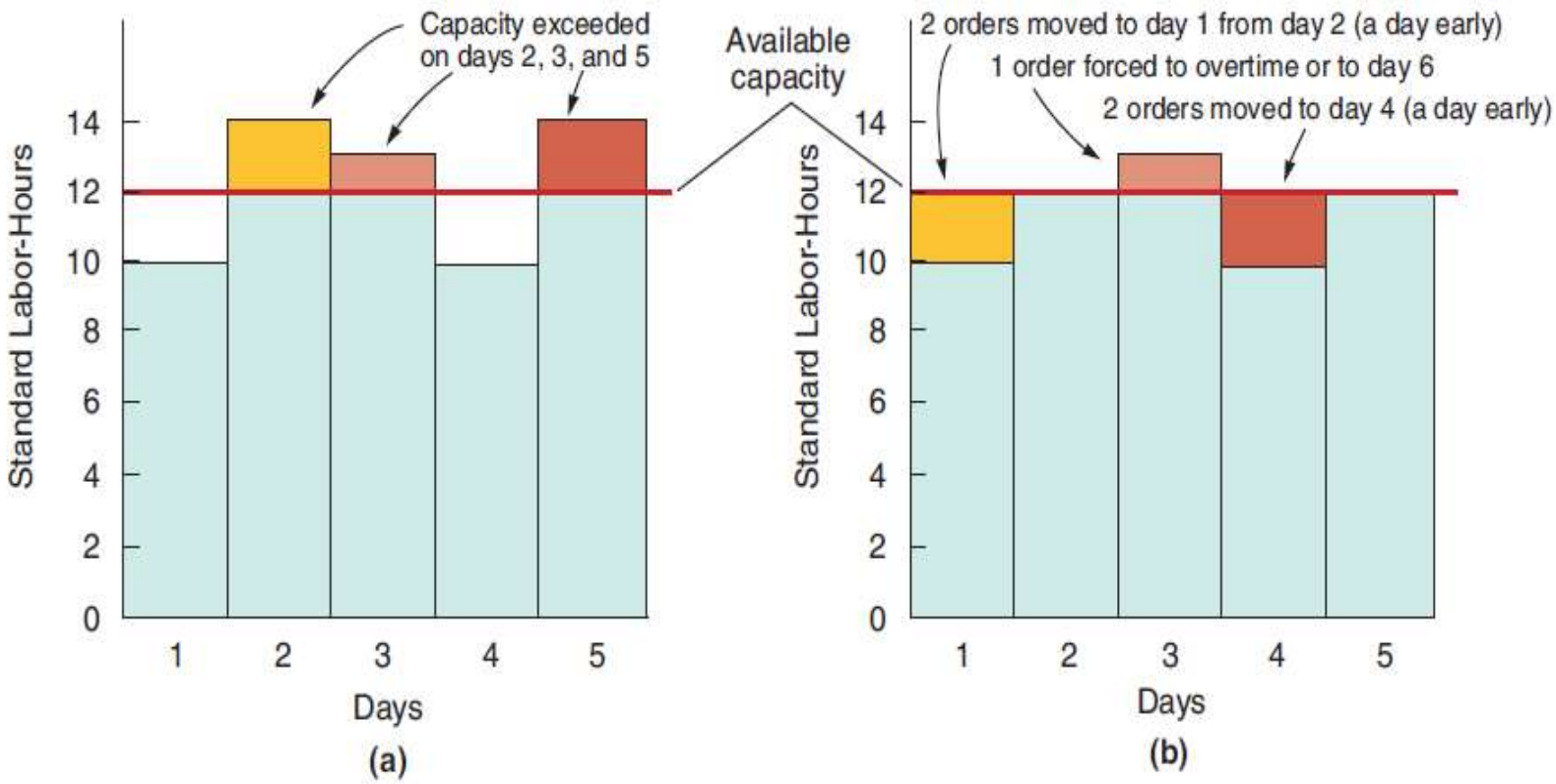


Source: Heizer, Render, Muckson

Smoothing Tactics

1. ***Overlapping***
   - Sends part of the work to following operations before the entire lot is complete
   - Reduces lead time
2. ***Operations splitting***
   - Sends the lot to two different machines for the same operation
   - Shorter throughput time but increased setup costs
3. ***Order or lot splitting***
   - Breaking up the order into smaller lots and running part earlier (or later) in the schedule

Class Exercise

- Kevin Watson, the production planner at Wiz Products, needs to develop a capacity plan for a work center.
- He has the production orders shown below for the next 5 days.
- There are 12 hours available in the work cell each day. The parts being produced require 1 hour each.

Day	1	2	3	4	5
Orders	10	14	13	10	14

Source: Heizer, Render, Muckson

Solution

SOLUTION ▶

DAY	UNITS ORDERED	CAPACITY REQUIRED (HOURS)	CAPACITY AVAILABLE (HOURS)	UTILIZATION: OVER/ (UNDER) (HOURS)	PRODUCTION PLANNER'S ACTION	NEW PRODUCTION SCHEDULE
1	10	10	12	(2)		12
2	14	14	12	2	Split order: move 2 units to day 1	12
3	13	13	12	1	Split order: move 1 unit to day 6 or request overtime	13
4	10	10	12	(2)		12
5	14	14	12	2	Split order: move 2 units to day 4	12
	61					

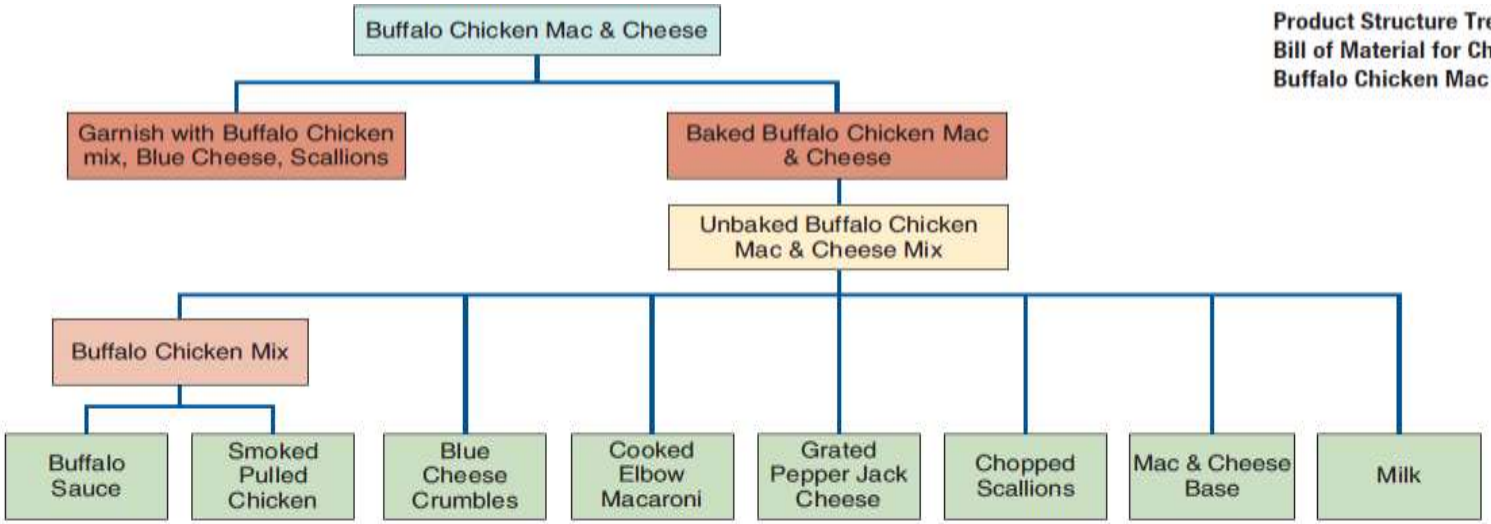
Source: Heizer, Render, Muckson

Any Question

MRP in Services

- Restaurants.
- Hospitals
- Hotels

Example



Product Structure Tree and Bill of Material for Chef John's Buffalo Chicken Mac & Cheese

(b) **BILL OF MATERIALS**

Production Specification	**Buffalo Chicken Mac & Cheese (6 portions)**				
Ingredients	*Quantity*	*Measure*	*Unit Cost*	*Total Cost*	*Labor Hrs.*
Elbow Macaroni (large, uncooked)	20.00	oz.	$ 0.09	$ 1.80	
Cheese—Pepper Jack (grated)	10.00	oz.	0.17	1.70	
Mac and Cheese Base (from refrigerator)	32.00	oz.	0.80	25.60	
Milk	4.00	oz.	0.03	0.12	
Smoked Pulled Chicken	2.00	lb.	2.90	5.80	
Buffalo Sauce	8.00	oz.	0.09	0.72	
Blue Cheese Crumbles	4.00	oz.	0.19	0.76	
Scallions	2.00	oz.	0.18	0.36	
Total Labor Hours					0.2 hrs

Source: Heizer, Render, Muckson

Distribution Resource Planning (DRP)

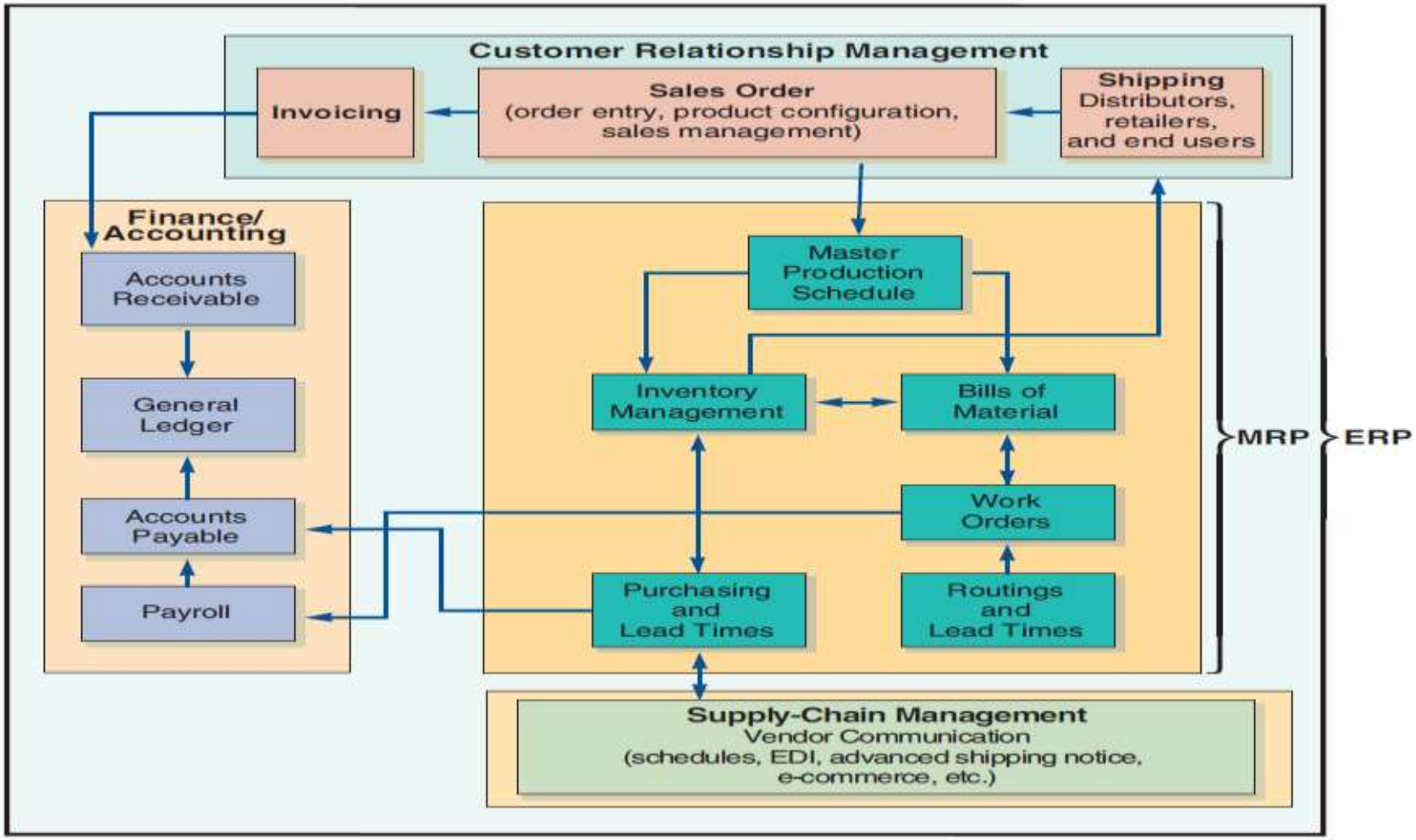
Using dependent demand techniques throughout the supply chain

- Expected demand or sales forecasts become gross requirements
- All other levels are computed
- DRP pulls inventory through the system
- Small and frequent replenishments

ERP

- Enterprise resource planning (ERP) : An information system for identifying and planning the enterprise wide resources needed to take, make, ship, and account for customer orders.
- Enterprise resource planning (ERP) is software that allows companies to (1) automate and integrate many of their business processes, (2) share a common database and business practices throughout the enterprise, and (3) produce information in real time.

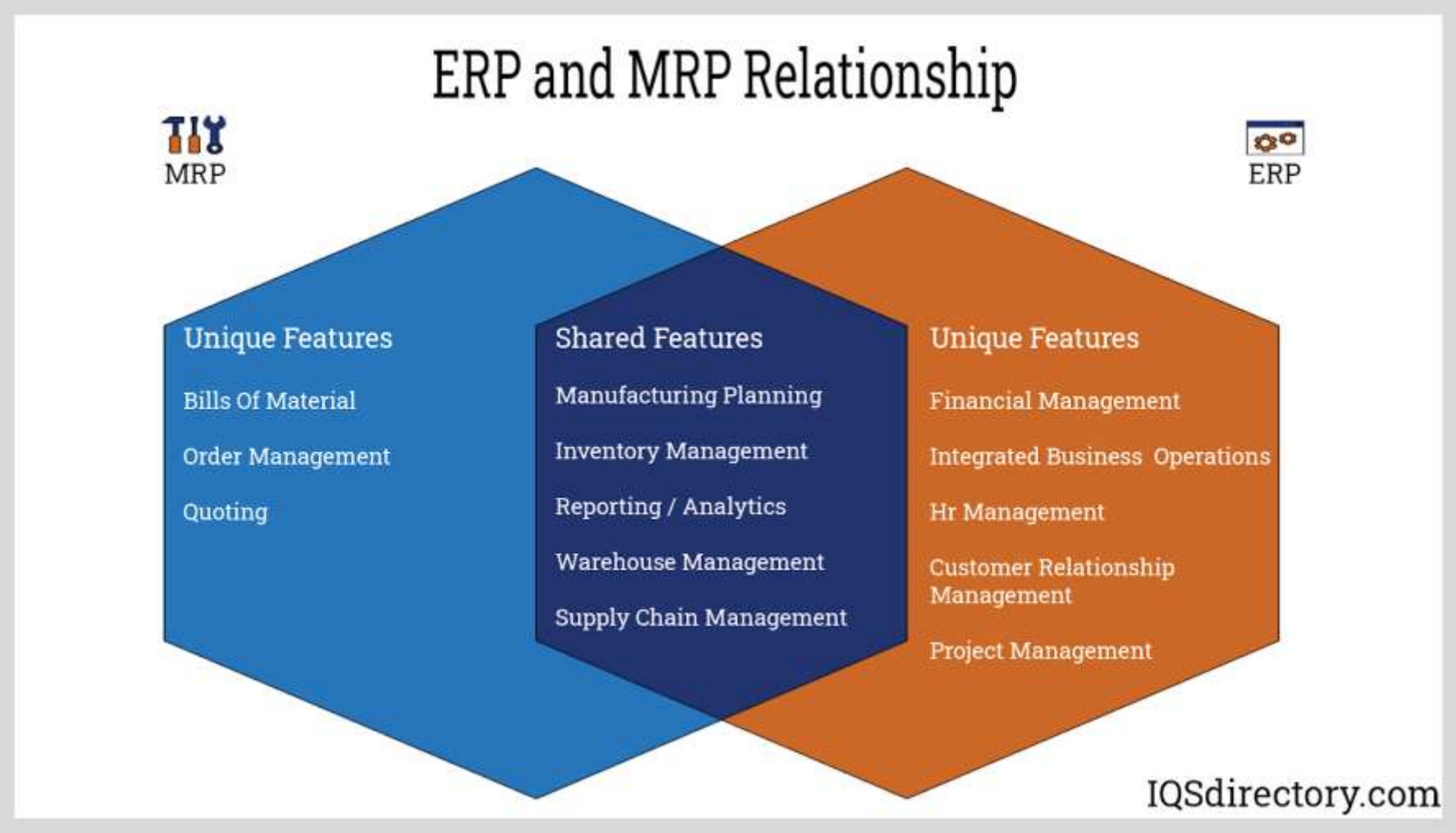
ERP



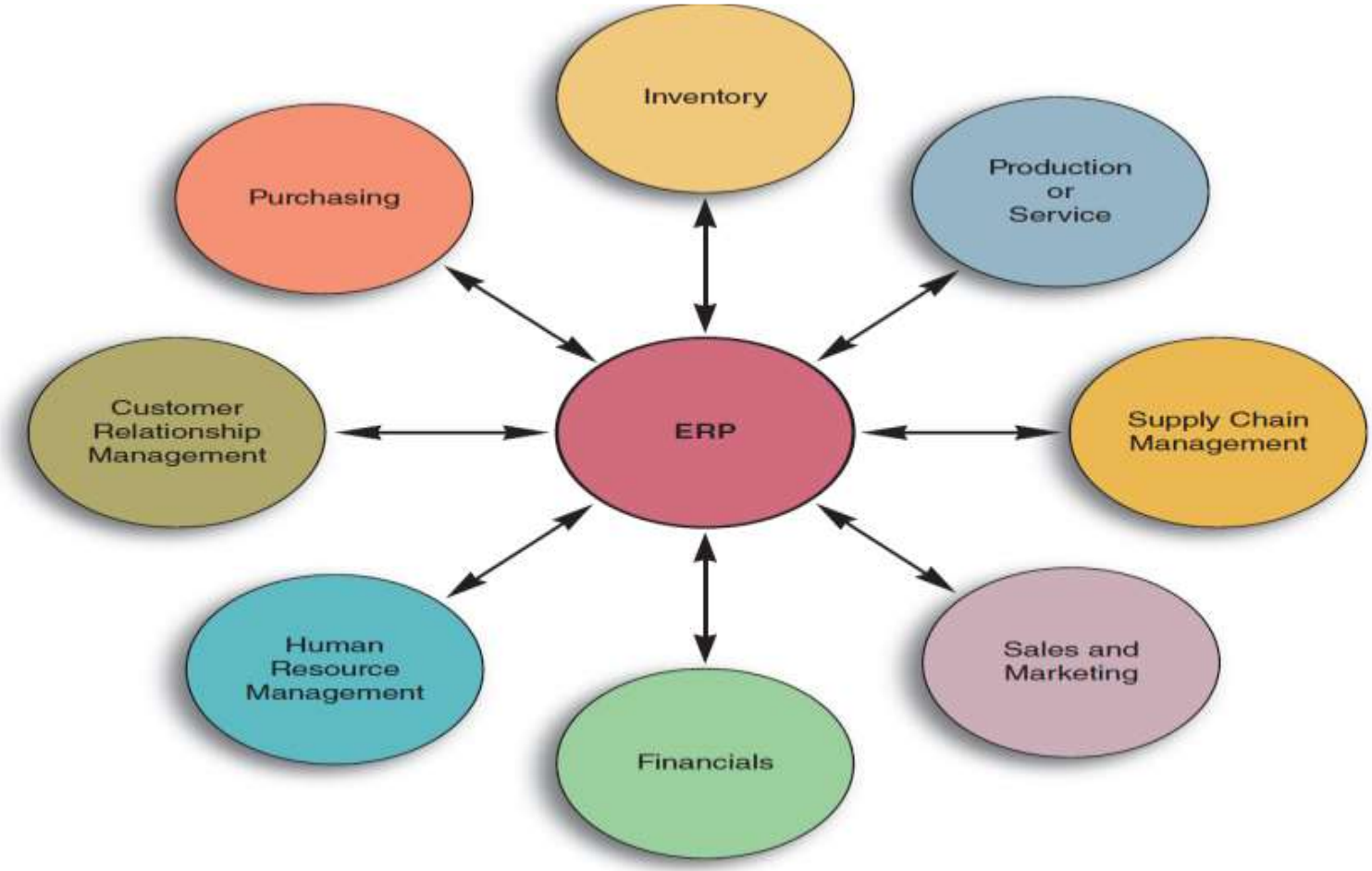
Source: Heizer, Render, Muckson

Material Requirements Planning (MRP) & ERP

Enterprise Resource Planning (ERP)



ERP



Source: Operation Management by William J Stevenson

Material Requirements Planning (MRP) & ERP

SAP's Modules for ERP

CASH TO CASH

Covers all financial related activity:

- Accounts receivable
- Accounts payable
- General ledger
- Treasury
- Cash management
- Asset management

PROMOTE TO DELIVER

Covers front-end customer-oriented activities:

- Marketing
- Quote and order processing
- Transportation
- Documentation and labeling
- After sales service
- Warranty and guarantees

DESIGN TO MANUFACTURE

Covers internal production activities:

- Design engineering
- Production engineering
- Plant maintenance
- Shop floor reporting
- Contract/project management
- Subcontractor management

RECRUIT TO RETIRE

Covers all HR- and payroll-oriented activity:

- Time and attendance
- Travel and expenses
- Payroll

PROCURE TO PAY

Covers sourcing activities:

- Vendor sourcing
- Purchase requisitioning
- Purchase ordering
- Purchase contracts
- Inbound logistics
- Supplier invoicing/matching
- Supplier payment/ settlement
- Supplier performance

DOCK TO DISPATCH

Covers internal inventory management:

- Warehousing
- Distribution planning
- Forecasting
- Replenishment planning
- Physical inventory
- Material handling

Figure 14.11

SAP's Modules for ERP

Enterprise Resource Planning (ERP)

- **ERP systems have the potential to**
  - Reduce transaction costs
  - Increase the speed and accuracy of information
- **Facilitates a strategic emphasis on JIT systems and supply chain integration**
- **Can be expensive and time-consuming to install**

ERP in the Service Sector

- ERP systems have been developed for health care, government, retail stores, hotels, and financial services
- Also called efficient consumer response (ECR) systems in the grocery industry
- Objective is to tie sales to buying, inventory, logistics, and production

Further Reading / References

1. Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 14
2. Operation Management by William J Stevenson—Chapter 12

Any Question

THANK YOU